

Question	Answer	Marks
3(a)(i)	mass	B1
3(a)(ii)	charge	B1
3(b)(i)	$E = V/d$ or $E = F/q$	C1
	$F = (1.2 \times 10^3 \times 4.2 \times 10^{-9}) / 3.6 \times 10^{-2}$	C1
	$= 1.4 \times 10^{-4} \text{ N}$	A1
3(b)(ii)	$W = mg$	C1
	$= 5.9 \times 10^{-6} \times 9.81$	C1
	resultant force $= 1.4 \times 10^{-4} - (5.9 \times 10^{-6} \times 9.81)$	
	$a = F/m$	C1
	$a = [1.4 \times 10^{-4} - (5.9 \times 10^{-6} \times 9.81)] / [5.9 \times 10^{-6}] = 14 \text{ ms}^{-2}$	A1
3(b)(iii)	1. $s = ut + \frac{1}{2}at^2$ $1.8 \times 10^{-2} = \frac{1}{2} \times 14 \times t^2$	C1
	$t = 0.051 \text{ s}$	A1
	2. $p = 0.75 \times 0.051$ $= 0.038 \text{ m}$	A1